

Software Requirements Specification (SRS) Template

Project: Task Manager Application

Version: 1.0

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Revision history

Version	Date	Author	Change summary
1.0	09-09-2026	Project Team	Initial SRS prepared with detailed requirements, security, diagrams and RTM

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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for a Task Manager Application. It defines the functional and non-functional requirements, external interfaces, security requirements, acceptance criteria, and verification references for the application. The document is intended to guide development, testing, evaluation, and future maintenance of the project.

1.2 Scope

The Task Manager Application covers creation, organization, tracking, updating, searching, prioritization, and completion of tasks. It supports user authentication, task details such as priority and due date, task status management, filtering, notifications, and task summaries. The application focuses on task-management activities and does not cover unrelated enterprise resource planning or external organizational systems.

1.3 Audience

Students and project developers, QA/testers, course coordinator/evaluators, system administrators, and end users who need to create and manage tasks.

1.4 Definitions

List of terms and acronyms: SRS, UI, API, CRUD, FR, NFR, RTM, OTP, ID, Admin, User.

2. Overall description

2.1 Product perspective

The Task Manager Application is a software system that provides a centralized interface for managing personal or assigned tasks. It maintains task information, allows users to update task progress, and provides views for pending and completed work. The application may interact with a persistent data store and notification services through defined software interfaces.

2.2 Major product functions (detailed)

- User registration and authentication
- Create, view, edit and delete tasks
- Assign priority and due dates to tasks
- Maintain task status such as Pending, In Progress and Completed
- Categorize and organize tasks
- Search and filter tasks
- Sort tasks by priority, due date or status
- Display pending and overdue tasks
- Provide task completion and progress information
- Generate a task summary
- Provide notifications or reminders for due tasks
- Support controlled access to user and administrative functions
- Maintain task and user records persistently
- Validate user input and task data
- Provide logout and session termination

2.3 User roles and characteristics (expanded)

- User: basic computer/application knowledge; expects simple navigation and quick task updates.
- Administrator/Manager: manages access and, where enabled, reviews overall task status and user records.
- Project Evaluator: reviews functionality, requirement compliance, test evidence and documentation.

2.4 Operating environment

The application is intended to operate on standard desktop or mobile computing environments through its supported user interface. It requires a functioning runtime environment, persistent data storage, and network connectivity where server-based services or notifications are used.

2.5 Constraints

- Only authenticated users shall access protected task data.
- Task records shall be associated with the correct user or authorized assignee.
- User input shall be validated before it is stored.
- The application shall follow the approved project architecture and available development environment.
- Security controls shall prevent unauthorized access to task and account information.

3. External interface requirements

3.1 User interfaces

Primary UI: a clear dashboard with navigation for tasks, task creation, task details, filters, search, notifications and account functions. Forms shall provide labels, validation messages and confirmation feedback. Important states such as overdue, completed and high-priority tasks shall be distinguishable through clear text and UI indicators.

3.2 Hardware interfaces

- Standard keyboard and pointing/touch input devices for task entry and navigation.
- Display capable of presenting the application UI and status messages.
- Optional notification-capable device where reminders are supported.

3.3 Software interfaces

- Application data store/database for users, tasks, categories, statuses and related records.
- Authentication/session component for login and access control.
- Notification service/interface for reminders where enabled.
- Optional API layer for communication between the client interface and application services.

3.4 Communications

Where the application uses networked services, communication shall use authenticated requests and secure transport. Failed requests shall produce a user-understandable error message, and the application shall avoid losing already saved task data because of a temporary communication failure.

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow TMA-F-###.

4.1 User Authentication and Account Management

Description: Authenticate users, manage sessions, and protect account access.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
TMA-F-001	The system shall allow a registered user to log in using valid credentials.	Functional	High	User	Valid credentials create an authenticated session. Test: TC-Auth-01	Requires user account store.
TMA-F-002	The system shall reject invalid login credentials and display an appropriate error message.	Functional	High	Security / User	Invalid credentials do not create a session. Test: TC-Auth-02	Authentication service required.
TMA-F-003	The system shall allow an authenticated user	Functional	High	User	After logout, protected pages cannot be	Session management required.

	to log out and terminate the active session.				accessed without re-authentication. Test: TC-Auth-03	
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4.2 Task Creation and Editing

Description: Create and maintain task records through controlled user actions.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
TMA-F-010	The system shall allow an authenticated user to create a task with a title and task details.	Functional	High	User	A valid task is stored and displayed in the user's task list. Test: TC-Task-01	Requires persistent storage.
TMA-F-011	The system shall allow the user to edit the details of an existing task to which the user has access.	Functional	High	User	Updated details are saved and displayed correctly. Test: TC-Task-02	Authorization required.
TMA-F-012	The system shall allow the user to delete an existing task after confirmation.	Functional	High	User	Confirmed deletion removes the task from the active task list. Test: TC-Task-03	Confirmation UI required.

4.3 Task Organization and Status

Description: Organize tasks using priority, due dates, categories and status.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
TMA-F-020	The system shall allow a user to assign a priority level to a task.	Functional	High	User	Selected priority is stored and displayed with the task. Test: TC-Org-01	Uses configured priority values.
TMA-F-021	The system shall allow a user to set or change the due date of a task.	Functional	High	User	Valid due date is stored and displayed. Test: TC-Org-02	Date validation required.
TMA-F-022	The system shall allow a user to change task status among the configured workflow states.	Functional	High	User	Status change is persisted and reflected in task views. Test: TC-Org-03	Status values must be configured.
TMA-F-023	The system shall allow tasks to be assigned to categories or labels.	Functional	Medium	User	Selected category/label appears in the task record and filters. Test: TC-Org-04	Category data required.

4.4 Search, Filter and Monitoring

Description: Help users locate tasks and monitor task progress and deadlines.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
TMA-F-030	The system shall allow users to search accessible tasks using task title or relevant task text.	Functional	High	User	Matching accessible tasks are returned. Test: TC-Search-01	Search index/query required.
TMA-F-031	The system shall allow users to filter tasks by status, priority, category and due-date condition.	Functional	High	User	Applying a filter returns only matching accessible tasks. Test: TC-Search-02	Filter fields required.
TMA-F-032	The system shall allow users to sort tasks by due date, priority or status.	Functional	Medium	User	Selected sort order is applied correctly. Test: TC-Search-03	Sorting must be deterministic.
TMA-F-033	The system shall identify tasks that are overdue based	Functional	High	User	An incomplete task past its due date is marked as overdue.	Requires valid date/time.

	on the current date and their due dates.				Test: TC-Monitor-01	
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4.5 Notifications, Summary and Administration

Description: Provide reminders, task summaries and controlled administrative functions.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
TMA-F-040	The system shall provide a reminder or notification for configured upcoming or overdue tasks where notifications are enabled.	Functional	Medium	User	Eligible task produces the configured notification. Test: TC-Notify-01	Notification service may be required.
TMA-F-041	The system shall display a summary of task counts by relevant status, such as pending, in-progress and completed.	Functional	Medium	User	Summary counts match the underlying accessible task records. Test: TC-Summary-01	Data aggregation required.
TMA-F-042	The system shall restrict administrative functions to authorized administrator/manager users.	Functional	High	Security / Admin	Unauthorized users cannot access admin functions. Test: TC-Admin-01	Role-based access control required.

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans. IDs follow TMA-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
TMA-NF-001	Common user actions shall respond within 3 seconds for at least 90% of transactions under normal project test load.	Performance	High	90th percentile response time ≤ 3 seconds. Test: TC-Perf-01
TMA-NF-002	The application shall maintain availability of at least 99% during the defined operational period, excluding planned maintenance.	Reliability	High	Availability report ≥ 99%. Test: TC-Rel-01
TMA-NF-003	The system shall protect user credentials and sensitive task data from unauthorized access and shall not expose credentials in application logs.	Security	High	Security test and log review pass. Test: TC-Sec-01
TMA-NF-004	The system shall preserve valid task records after successful save operations and shall recover gracefully from an application restart.	Data Integrity / Reliability	High	Saved records remain unchanged after restart. Test: TC-Data-01
TMA-NF-005	The user interface shall provide readable labels, consistent navigation, clear validation messages and keyboard/touch accessible controls.	Usability / Accessibility	Medium	Usability checklist and accessibility review pass. Test: TC-UX-01

5.1 Security

5.1.1 Security Objectives

- SO-01: Ensure that only authenticated and authorized users can access task and account information.
- SO-02: Preserve confidentiality and integrity of credentials and task data during storage, processing and communication.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
TMA-SR-001	The system shall require authentication before allowing access to protected task data.	Security	High	Unauthenticated access is blocked. Test: TC-Sec-01
TMA-SR-002	The system shall enforce role-based authorization for administrative functions.	Security	High	Non-admin user is denied admin access. Test: TC-Sec-02
TMA-SR-003	The system shall protect passwords using secure password handling and shall never store passwords in plaintext.	Security	High	Password storage review confirms secure representation. Test: TC-Sec-03
TMA-SR-004	The system shall use secure transport for network communication whenever credentials or task data are transmitted.	Security	High	Network/security test confirms protected transport. Test: TC-Sec-04
TMA-SR-005	The system shall validate and sanitize user-provided input before processing or storing it.	Security	High	Invalid/malicious test inputs are rejected or safely handled. Test: TC-Sec-05

6. Quality attributes & Acceptance tests

Exit criteria for acceptance: All high-priority functional requirements shall be implemented and verified, no critical NFR or security failures shall remain, and the RTM shall show coverage for the defined test cases.

Acceptance test suites

- Authentication and session tests: TC-Auth-01 to TC-Auth-03
- Task CRUD tests: TC-Task-01 to TC-Task-03

- Task organization tests: TC-Org-01 to TC-Org-04
- Search, filtering and monitoring tests: TC-Search-01 to TC-Search-03 and TC-Monitor-01
- Notification and summary tests: TC-Notify-01 and TC-Summary-01
- Administration and security tests: TC-Admin-01 and TC-Sec-01 to TC-Sec-05
- Performance, reliability, data integrity and usability tests: TC-Perf-01, TC-Rel-01, TC-Data-01 and TC-UX-01

7. System models and diagrams

7.1 UML Use-Case diagram

At least 2 UML use-case diagrams are included to represent the primary user interactions and administration/monitoring interactions.

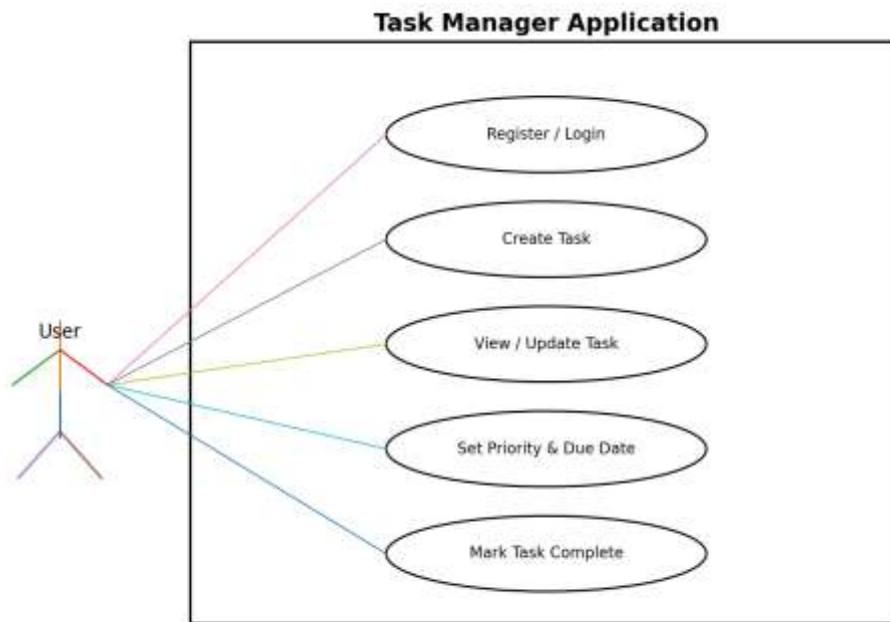


Figure 1. UML Use-Case Diagram: User Task Management

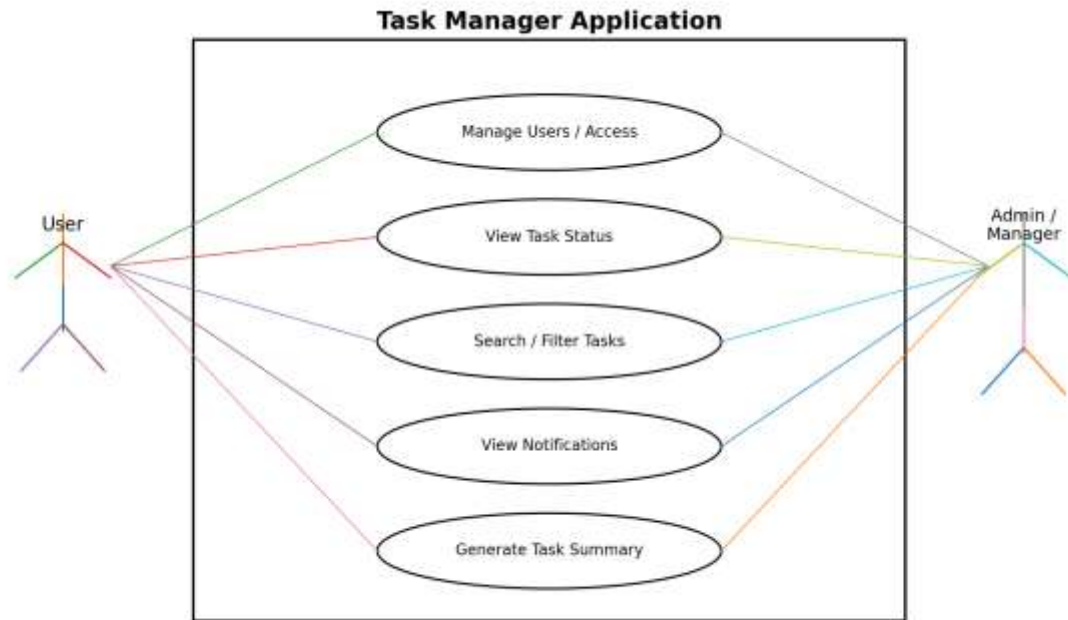


Figure 2. UML Use-Case Diagram: Monitoring and Administration

8. Requirements Traceability Matrix (RTM)

The RTM maps selected requirements to design references, modules, and test cases. Status values use N/P/A: Not Started / Partially Implemented / Approved.

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
TMA-F-001	User login	4.1 / DS-Auth-01	Auth Module	TC-Auth-01	N	
TMA-F-010	Create task	4.2 / DS-Task-01	Task Module	TC-Task-01	N	
TMA-F-011	Edit task	4.2 / DS-Task-02	Task Module	TC-Task-02	N	
TMA-F-012	Delete task	4.2 / DS-Task-03	Task Module	TC-Task-03	N	
TMA-F-020	Set priority	4.3 / DS-Org-01	Task Module	TC-Org-01	N	
TMA-F-022	Update status	4.3 / DS-Org-03	Task Module	TC-Org-03	N	
TMA-F-030	Search tasks	4.4 / DS-Search-01	Search Module	TC-Search-01	N	
TMA-F-031	Filter tasks	4.4 / DS-Search-02	Search Module	TC-Search-02	N	
TMA-F-033	Overdue detection	4.4 / DS-Monitor-01	Monitoring Module	TC-Monitor-01	N	
TMA-F-040	Task reminders	4.5 / DS-Notify-01	Notification Module	TC-Notify-01	N	
TMA-F-042	Admin access	4.5 / DS-Admin-01	Admin Module	TC-Admin-01	N	
TMA-NF-001	Response time target	5 / DS-Perf-01	Application Services	TC-Perf-01	N	
TMA-NF-003	Security protection	5 / DS-Sec-01	Security Layer	TC-Sec-01	N	
TMA-SR-002	Role authorization	5.1.2 / DS-AuthZ-01	Access Control	TC-Sec-02	N	
TMA-SR-003	Password protection	5.1.2 / DS-Auth-02	Auth Module	TC-Sec-03	N	

Note: This SRS follows the organization, terminology, requirement style, acceptance-criteria approach, and RTM structure of the supplied reference template. Technology-specific details not provided by the team have intentionally been kept technology-neutral.